

SUPERELLIPTIC SELMER GROUPS

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1. THE PROBLEMS

Definition 1.0.1. Let k be a field. (♠ TODO: projective closure, plane curve, affine equation) (♠ TODO: smooth completion of an affine curve) A *superelliptic curve over k* is the smooth completion of the plane curve with affine equation of the form

$$y^d = f(x)$$

where $f(x) \in k[x]$ and $d \geq 2$ is some integer.

Definition 1.0.2. Let k be a field. Let C be a superelliptic curve (Definition 1.0.1) over k given as the smooth completion of the plane curve with affine equation

$$y^d = f(x)$$

where $f(x) \in k[x]$ and $d \geq 2$ is some integer. Given an element $u \in k^\times$, the *μ_d -twist of C* is the superelliptic curve given as the projective closure of the plane curve with affine equation

$$y^d = \frac{1}{u} f(x).$$

Here is the main problem:

Context 1.0.3. Let $K = \mathbb{F}_q(t)$ be a global function field of characteristic coprime to 6. Let $\ell \geq 5$ be a prime number such that $\gcd(\ell, q) = 1$. Write $F_{n,\ell}$ for the set of ℓ th power free polynomials over $\mathbb{F}_q$ of degree n . Let v be a prime number coprime to $6q\ell$.

Context 1.0.4. Assume Context 1.0.3. Fix a superelliptic curve (Definition 1.0.1) C/K whose affine equation is given by

$$C : y^\ell = f(x)$$

for some non-trivial polynomial $f(x) \in K[x]$.

Given an ℓ -th power free polynomial $g \in \mathbb{F}_q[t]$ write C_g for the μ_ℓ -twist (Definition 1.0.2) of C by g . In other words C_g is the smooth completion of the plane curve with affine equation of the form

$$C_g : gy^\ell = f(x)$$

Problem 1.0.5. Assume Context 1.0.3 and Context 1.0.4

For each integer $j \geq 0$, what is the asymptotic value of the quantity

$$\frac{\#\{g \in F_{n,\ell} : \dim_{\mathbb{F}_v} \text{Sel}_v(\text{Jac}(C_g)/K) = j\}}{\#F_{n,\ell}}$$

(Definition D.0.2) (Definition A.1.3) as $n \rightarrow \infty$?

Here are subproblems

Problem 1.0.6.

Problem 1.0.7. Assume Context 1.0.3.

Write $\mathcal{J}_g$ for the Néron model of $\text{Jac}(C_g)$ over

2. HALL'S BIG MONODROMY APPROACH

Theorem 2.0.1 (See e.g. [?, Chapter V Corollary 16.4]). Let $\mathbb{F}_q$ be a finite field of characteristic p with q elements, let G be a connected algebraic group over $\mathbb{F}_q$, and let F be the relative q th power Frobenius endomorphism on $G/\mathbb{F}_q$. The Lang map associated to F is a separable Galois étale (surjective) isogeny whose kernel is precisely $G(\mathbb{F}_q)$, the group of $\mathbb{F}_q$ -points of G . (♠ TODO: Borel's book states the Lang isogeny as separable, and surjective, but I don't know what shows the Galoisness and étaleness)

Definition 2.0.2. Let $G/\mathbb{F}_q$ be a connected commutative algebraic group. Let $\ell \neq \text{Char } \mathbb{F}_q$ be a prime number. Let $n \geq 1$. By Lang's theorem (Theorem 2.0.1), given a connected commutative algebraic group $G/\mathbb{F}_q$, we have we have a short exact sequence

$$1 \rightarrow G(\mathbb{F}_{q^n}) \rightarrow G_{\mathbb{F}_{q^n}} = G \times_{\mathbb{F}_q} \mathbb{F}_{q^n} \xrightarrow{L} G_{\mathbb{F}_{q^n}} \rightarrow 1$$

where L is the Lang isogeny. Since this identifies $G_{\mathbb{F}_{q^n}}$ with a finite étale $G(\mathbb{F}_{q^n})$ -cover of $G_{\mathbb{F}_{q^n}}$, we have a corresponding surjective group homomorphism $\pi_1^{\text{ét}}(G_{\mathbb{F}_{q^n}}, e) \rightarrow G(\mathbb{F}_{q^n})$.

1. Let $\chi : G(\mathbb{F}_{q^n}) \rightarrow \mathbb{F}_\ell^\times$ be a character. There is a lisse sheaf $\mathcal{L}_\chi$ of rank 1 with coefficients in $\mathbb{F}_\ell$ on $G_{\mathbb{F}_{q^n}} = G \times_{\mathbb{F}_q} \mathbb{F}_{q^n}$ corresponding (Theorem E.0.2) to the composed group homomorphism

$$\pi_1^{\text{ét}}(G_{\mathbb{F}_{q^n}}, e) \rightarrow G(\mathbb{F}_{q^n}) \xrightarrow{\chi} \mathbb{F}_\ell^\times.$$

We may call this lisse sheaf the *Kummer sheaf associated to χ* .

Similarly, Kummer sheaves are associated to characters $\chi : G(\mathbb{F}_{q^n}) \rightarrow \mathbb{Z}_\ell^\times$, $\chi : G(\mathbb{F}_{q^n}) \rightarrow \mathbb{Q}_\ell^\times$, $\chi : G(\mathbb{F}_{q^n}) \rightarrow \overline{\mathbb{Q}}_\ell^\times$, etc.; each type of Kummer sheaf is a lisse sheaf on $G_{\mathbb{F}_{q^n}}$ with appropriate coefficients.

Definition 2.0.3. (♠ TODO: For the statements that depend on this one, it must be considered whether the points are codimension 1; I had not written that discussion down before — the well-definedness up to conjugacy is true only for codimension 1 points) Let X be an integral normal scheme with generic point η and function field K . Let $\bar{\eta}$ be a geometric point of X whose image is η . Let $U \subseteq X$ be a non-empty open subscheme. Let $x \in X \setminus U$ be a point. Let v be a valuation of K centered at x .

Recall that there is a natural surjection

$$\text{Gal}(K^{\text{sep}}/K) \twoheadrightarrow \pi_1^{\text{ét}}(U, \bar{\eta})$$

(Proposition E.0.5)

1. The *decomposition subgroup* D_v of $\pi_1^{\text{ét}}(U, \bar{\eta})$ is the image of the decomposition subgroup (Definition E.0.3) $D_{K,v}$ of $\text{Gal}(K^{\text{sep}}/K)$ in $\pi_1^{\text{ét}}(U, \bar{\eta})$.
2. The *inertia subgroup* I_v of $\pi_1^{\text{ét}}(U, \bar{\eta})$ is the image of the inertia subgroup (Definition E.0.4) $I_{K,v}$ of $\text{Gal}(K^{\text{sep}}/K)$ in $\pi_1^{\text{ét}}(U, \bar{\eta})$.

In the case that x is a point of codimension 1 in X , given two choices v_1 and v_2 of valuations centered at x , the groups D_{v_1} and D_{v_2} and the groups I_{v_1} and I_{v_2} are conjugate to each other in $\pi_1^{\text{ét}}(U, \bar{\eta})$. As such, one might write D_x and I_x for the conjugacy classes.

Definition 2.0.4. Let X be an connected normal scheme with generic point η and function field K . Let $\bar{\eta}$ be a geometric point of X whose image is η . Let $U \subseteq X$ be a non-empty open subscheme. Let $x \in X \setminus U$ be a point.

Let $\mathcal{F}$ be one of the following:

1. A finite locally constant sheaf of sets on $U_{\text{ét}}$
2. A finite locally constant sheaf of finite cardinality abelian groups on $U_{\text{ét}}$
3. A locally constant sheaf of finitely generated modules over a finite ring on $U_{\text{ét}}$
4. In the case that U is irreducible and geometrically unibranch, a finite type, locally constant sheaf of Λ -modules on $U_{\text{ét}}$
5. Let (R, λ) be a complete DVR such that R/λ^n is finite for all $n \geq 1$. Let E be the fraction field of R .
 - (a) A lisse object of $\text{Sh}(U, R)$ (??)
 - (b) A lisse object of $\text{Sh}(U, E)$ (??)
 - (c) For an algebraic extension K of E , a lisse object of $\text{Sh}(U, K)$ (??).

In each of the above cases for $\mathcal{F}$, there is a corresponding action of $\pi_1^{\text{ét}}(U, \bar{\eta})$ on the stalk $\mathcal{F}_{\bar{\eta}}$ (Theorem E.0.1) (Theorem E.0.2):

$$\pi_1^{\text{ét}}(U, \bar{\eta}) \rightarrow \text{Aut}(\mathcal{F}_{\bar{\eta}})$$

where $\text{Aut}(\mathcal{F}_{\bar{\eta}})$ is the automorphism group for the appropriate category that $\mathcal{F}_{\bar{\eta}}$ belongs to. Further recall that there is an inertia subgroup (Definition 2.0.3) I_x of $\pi_1^{\text{ét}}(U, \bar{\eta})$, well defined up to conjugation.

We say that $\mathcal{F}$ is *unramified at x* if the action of I_x on $\mathcal{F}_{\bar{\eta}}$ is trivial, i.e. the composed homomorphism

$$I_x \hookrightarrow \pi_1^{\text{ét}}(U, \bar{\eta}) \rightarrow \text{Aut}(\mathcal{F}_{\bar{\eta}})$$

is trivial. Otherwise, we say that $\mathcal{F}$ is *ramified at x* .

Let $C/\mathbb{F}_q$ be a smooth, geometrically connected curve. Let $X/K(C)$ be a superelliptic curve with affine model $y^r = f(x)$ for a polynomial $f(x) \in K(C)[x]$. Given $g \in K(C)^\times$, let X_g be the twist of X with affine model $g \cdot y^r = f(x)$. Write $\mathcal{X}_g \rightarrow C$ for the Néron model of X_g . For a prime number v not dividing $\text{Char } \mathbb{F}_q$, $\text{Pic}_{\mathcal{X}_g/C}^0[v]$ is a quasi-finite étale group scheme over C (♠ TODO: cite).

Context 2.0.5. Let q be a prime power. Fix an elliptic curve $E_1/\mathbb{F}_q(t)$ with non-constant j -invariant and let $\mathcal{E}_1 \rightarrow \mathbb{P}_{\mathbb{F}_q}^1$ be its Néron model (Definition B.0.4). Let ℓ be a prime not dividing q . The kernel $\mathcal{E}_1[\ell]$ is a étale group scheme over C and in particular is quasi-finite.

For a positive integer d , and for an algebraically closed extension field $\bar{k}$ of $\mathbb{F}_q$, identify a geometric point $(C_0, \dots, C_{d-1})(\text{Spec } \mathbb{F}_q[c_0, \dots, c_{d-1}])(\bar{k})$ for $(C_0, \dots, C_{d-1}) \in \bar{k}^d$ with a monic polynomial $t^d + C_{d-1}t^{d-1} + \dots + C_1t + C_0 \in \bar{\mathbb{F}}_q[t]$. Fixing a polynomial $m \in \mathbb{F}_q[t]$, let $F_d \subset \text{Spec } \mathbb{F}_q[c_0, \dots, c_{d-1}]$ be the parameter space of square-free polynomials of degree d relatively prime to m . Explicitly,

$$F_d = \text{Spec } \mathbb{F}_q[c_0, \dots, c_{d-1}] \setminus (\Delta \cup V(m))$$

(♠ TODO: resultant) where Δ is the discriminant locus in $\text{Spec } \mathbb{F}_q[c_0, \dots, c_{d-1}]$ and $V(m)$ is the locus of polynomials that are not coprime with m . For each geometric point of $F_d(\bar{\mathbb{F}}_q)$ corresponding to a polynomial $f \in \bar{\mathbb{F}}_q[t]$, let E_f/K be the quadratic twist of E_1 by f . Let $\mathcal{E}_f \rightarrow \mathbb{P}_{\mathbb{F}_q}^1$ be its Néron model (Definition B.0.4).

Let $S := \mathbb{P}_{\mathbb{F}_q}^1 \times F_d$; in accordance to the notations above, it has coordinates (t, c) . Let $P(t, c) = t^d + c_{d-1}t^{d-1} + \dots + c_1t + c_0 \in \mathbb{F}_q[t][c_0, \dots, c_{d-1}]$ be the universal monic polynomial of degree d over F_d — it is a function on $\mathbb{A}^1 \times F_d \subset S$. Let $\mathcal{E}_{\text{univ}} = \mathcal{E}_1 \times_{\mathbb{P}_{\mathbb{F}_q}^1} S \rightarrow S$. Let $\tilde{S} \subset \mathbb{A}_{\mathbb{F}_q}^1 \times S$, with coordinates (y, t, c) , be the closed subscheme defined by

$$w^2 = P(t, c).$$

The natural projection

$$\sigma : \tilde{S} \rightarrow S, \quad (w, t, c) \mapsto (t, c)$$

is a finite flat double cover. Let $U_{\text{ram}} \subset S$ be the branch locus of σ , explicitly given by $P(t, c) = 0$. Let $U = S \setminus U_{\text{ram}}$. In particular, the restriction $\sigma|_{\tilde{S}|_U} : \tilde{S}|_U \rightarrow U$ is a finite étale Galois cover with Galois group $\mathbb{Z}/2\mathbb{Z}$.

Let $\iota : \tilde{S} \rightarrow \tilde{S}$ be the involution $(w, t, c) \mapsto (-w, t, c)$. Writing (Q, w, t, c) for a point of

$$\sigma^* \mathcal{E}_{\text{univ}} = \mathcal{E}_1 \times_S \tilde{S} = (\mathcal{E}_1 \times_{\mathbb{P}_{\mathbb{F}_q}^1} S) \times_S \tilde{S} = \tilde{E}_1 \times_{\mathbb{P}_{\mathbb{F}_q}^1} \tilde{S},$$

there is an action by $\mathbb{Z}/2\mathbb{Z}$ on $\sigma^* \mathcal{E}_{\text{univ}}$ by the action

$$(Q, w, t, c) \mapsto (-Q, \iota(w, t, c)) = (-Q, -w, t, c).$$

This action is étale and free on the generic fiber. The quotient $\sigma^* \mathcal{E}_{\text{univ}} / (\mathbb{Z}/2\mathbb{Z})$ of $\sigma^* \mathcal{E}_{\text{univ}}$ by this action, which exists by faithfully flat descent, is a family of generalized elliptic curves S . (♠ TODO: generalized elliptic curve) (♠ TODO: understand how faithfully flat descent works)

Proposition 2.0.6. Assume Context 2.0.5.

(♠ TODO: should I be expressing this as a fiber of a point or a fiber over $\mathbb{P}_{\mathbb{F}_q}^1 \times \text{point}$? Because depending on whether I'm going for a point or a closed locus, I would be taking about a quadratic twist of E_1 or $\mathcal{E}_1$.) For every geometric point $\bar{f} \in F_d(\overline{\mathbb{F}_q})$ corresponding to a square-free polynomial $f(t) \in \overline{\mathbb{F}_q}[t]$, the restriction of $\mathcal{G} := (\sigma^* \mathcal{E}_{\text{univ}}) / (\mathbb{Z}/2\mathbb{Z})$, which is a family of generalized elliptic curves over S , to the fiber $\mathbb{P}_{\overline{\mathbb{F}_q}}^1 \times \{\bar{f}\} \subset S$ is isomorphic to the Néron model $\mathcal{E}_f \rightarrow \mathbb{P}_{\overline{\mathbb{F}_q}}^1$ of the quadratic twist E_f . (♠ TODO: recheck proof, evaluating why it's the smooth locus)

Proof. Since 2 is invertible in $\mathbb{F}_q$, the Néron model $\mathcal{E}_1 \rightarrow C$ admits a relative Weierstrass equation

$$y^2 = x^3 + Ax + B$$

(♠ TODO: probably a relative Riemann-Roch theorem yields this) locally over C where A and B are sections of $\omega_{\mathcal{E}_1/C}^{\oplus 4}$ and $\omega_{\mathcal{E}_1/C}^{\oplus 6}$ respectively. Write $\pi : S = \mathbb{P}_{\mathbb{F}_q}^1 \times F_d \rightarrow F_d$ be the projection and let $\sigma : \tilde{S} \rightarrow S$ be the double cover as in Context 2.0.5. The pullback of $\mathcal{E}_1 \rightarrow C$ via $\sigma \circ p$ is the total space $\sigma^* \mathcal{E}_{\text{univ}} \rightarrow \tilde{S}$, which has equation

$$y^2 = x^3 + (\sigma \circ \pi)^* A \cdot x + (\sigma \circ \pi)^* B.$$

The involution $(Q, w, t, c) \mapsto (-Q, -w, t, c)$ on $\sigma^* \mathcal{E}_{\text{univ}}$ fixes x and negates y and w . Setting

$$X = x, \quad Y = wy,$$

these are invariant under the action since $(-w)(-y) = wy$. These satisfy

$$Y^2 = w^2 y^2 = P(t, c) \cdot (X^3 + (\sigma \circ \pi)^* A \cdot X + (\sigma \circ \pi)^* B),$$

which is an equation that descends to S because $P(t, c)$, $(\sigma \circ \pi)^* A$, and $(\sigma \circ \pi)^* B$ do not depend on w . In other words, we have a family of elliptic curves over S with Weierstrass equation given by

$$Y^2 = P(t, c) \cdot (X^3 + p^* A \cdot X + p^* B).$$

The fibers of this relative generalized elliptic curve over $(t_0, C) \in S(\mathbb{F}_q)$ is thus indeed the quadratic twist of $\mathcal{E}_1$ by $P(t_0, c_0)$. $\square$

Proposition 2.0.7. Assume Context 2.0.5. Let $\mathcal{G} = \sigma^* \mathcal{E}_{\text{univ}}/(\mathbb{Z}/2\mathbb{Z})$, which is a family of generalized elliptic curves over S . (♠ TODO: semistable) Let $j : S_{\text{sm}} \hookrightarrow S$ be the open immersion of the semistable locus, i.e. the locus where $\mathcal{G} \rightarrow S$ has good or multiplicative reduction. The kernel $\mathcal{G}[\ell]$ of multiplication by ℓ is an étale group scheme over S that restricts to a finite group scheme over S_{sm} . We identify $\mathcal{G}[\ell]$ with a an étale $\mathbb{F}_\ell$ -sheaf $\mathcal{G}_\ell$ on S whose restriction to S_{sm} is lisse. In particular, $j_* j^* \mathcal{G}_\ell$ is a constructible $\mathbb{F}_\ell$ -sheaf on S . Let $\pi : S = \mathbb{P}_{\mathbb{F}_q}^1 \times F_d \rightarrow F_d$ be the projection. The first higher direct image

$$\mathcal{T}_{d,\ell} = R^1 \pi_*(j_* j^* \mathcal{G}_\ell)$$

is a lisse $\mathbb{F}_\ell$ -sheaf on F_d and for every geometric point $\bar{f} \in F_d(\overline{\mathbb{F}}_q)$, the geometric fiber is

$$\mathcal{T}_{d,\ell}|_{\bar{f}} \cong H^1(\mathbb{P}_{\overline{\mathbb{F}}_q}^1, \mathcal{E}_{f,\ell})$$

where $\mathcal{E}_{f,\ell}$ is the Néron model of ℓ -torsion subgroup $E_f[\ell]$ for the quadratic twist $E_f/\mathbb{F}_q(t)$.

Proof. We have a Cartesian diagram:

$$\begin{array}{ccc} \mathbb{P}_{\mathbb{F}_q}^1 & \xrightarrow{i_{\bar{f}}} & S = \mathbb{P}_{\mathbb{F}_q}^1 \times F_d \\ \downarrow \pi_{\bar{f}} & & \downarrow \pi \\ \text{Spec } \overline{\mathbb{F}}_q & \xrightarrow{\bar{f}} & F_d \end{array}$$

Since $\pi : S \rightarrow F_d$ is a proper morphism and $j_* j^* \mathcal{G}_\ell$ is a constructible $\mathbb{F}_\ell$ -sheaf on S , the Proper Base Change Theorem yields a canonical isomorphism: (♠ TODO: proper base change theorem in this context)

$$\mathcal{T}_{d,\ell}|_{\bar{f}} = (R^1 \pi_*(j_* j^* \mathcal{G}_\ell))_{\bar{f}} \cong H^1(\mathbb{P}_{\overline{\mathbb{F}}_q}^1, i_{\bar{f}}^* j_* j^* \mathcal{G}_\ell).$$

Let $j_{\bar{f}} : S_{\text{sm}} \times_S \mathbb{P}_{\overline{\mathbb{F}}_q}^1 \hookrightarrow \mathbb{P}_{\overline{\mathbb{F}}_q}^1$ denote the pullback of the open immersion j via the inclusion $i_{\bar{f}}$:

$$\begin{array}{ccc} S_{\text{sm}} \times_S \mathbb{P}_{\overline{\mathbb{F}}_q}^1 & \xrightarrow{i_{\bar{f},\text{sm}}} & S_{\text{sm}} \\ \downarrow j_{\bar{f}} & & \downarrow j \\ \mathbb{P}_{\overline{\mathbb{F}}_q}^1 & \xrightarrow{i_{\bar{f}}} & S. \end{array}$$

By the base change theorem for direct images of unramified torsion sheaves along open immersions, we have a canonical isomorphism:

$$i_{\bar{f}}^* j_* j^* \mathcal{G}_\ell \cong (j_{\bar{f}})_* i_{\bar{f},\text{sm}}^* j^* \mathcal{G}_\ell.$$

By Proposition 2.0.6, the restriction of $\mathcal{G}$ to the fiber $\mathbb{P}_{\overline{\mathbb{F}}_q}^1$ is canonically isomorphic to the Néron model $\mathcal{E}_f$. Since the formation of ℓ -torsion kernels commutes with base change, we

have the following sequence of canonical isomorphisms of group schemes over the smooth locus of the fiber:

$$i_{\bar{f}, \text{sm}}^* (\mathcal{G}|_{S_{\text{sm}}}[\ell]) \cong \left(i_{\bar{f}, \text{sm}}^* \mathcal{G}|_{S_{\text{sm}}} \right) [\ell] \cong \left(j_{\bar{f}}^* \mathcal{E}_f^{\text{sm}} \right) [\ell] \cong j_{\bar{f}}^* (\mathcal{E}_f^{\text{sm}}[\ell]).$$

Passing to the associated étale $\mathbb{F}_\ell$ -sheaves yields the identification $i_{\bar{f}, \text{sm}}^* j^* \mathcal{G}_\ell \cong j_{\bar{f}}^* \mathcal{E}_{f, \ell}$. Substituting this into the above base change isomorphism gives:

$$i_{\bar{f}}^* j_* j^* \mathcal{G}_\ell \cong (j_{\bar{f}})_* j_{\bar{f}}^* \mathcal{E}_{f, \ell}.$$

Since $\ell \nmid q$, the ℓ -torsion subgroup scheme of a Néron model satisfies the extension property $\mathcal{E}_{f, \ell} \cong (j_{\bar{f}})_* j_{\bar{f}}^* \mathcal{E}_{f, \ell}$. We thereby obtain the desired identification of the geometric fiber:

$$\mathcal{T}_{d, \ell}|_{\bar{f}} \cong H^1(\mathbb{P}_{\mathbb{F}_q}^1, \mathcal{E}_{f, \ell}).$$

(♠ TODO: verify the below) It remains to show that $\mathcal{T}_{d, \ell}$ is a lisse $\mathbb{F}_\ell$ -sheaf on F_d . Since F_d is smooth over $\mathbb{F}_q$, it suffices to show that the dimension of the geometric fibers $\dim_{\mathbb{F}_\ell} H^1(\mathbb{P}_{\mathbb{F}_q}^1, \mathcal{E}_{f, \ell})$ is constant for all $\bar{f} \in F_d(\overline{\mathbb{F}_q})$ and that the sheaf is unramified.

The dimension of $H^1(\mathbb{P}_{\mathbb{F}_q}^1, \mathcal{E}_{f, \ell})$ is given by the Grothendieck–Ogg–Shafarevich formula applied to the elliptic curve $E_f/\overline{\mathbb{F}_q}(t)$:

$$\dim_{\mathbb{F}_\ell} H^1(\mathbb{P}_{\mathbb{F}_q}^1, \mathcal{E}_{f, \ell}) = 2 \cdot \deg(\text{Cond}(E_f)) - 4 + \sum_{v \in \mathbb{P}^1} \delta_v(E_f[\ell]).$$

Because $\ell \neq p$, the wild ramification terms vanish ($\delta_v = 0$). Furthermore, since f is a square-free polynomial of degree d coprime to the conductor of E_1 , the degree of the conductor divisor $\deg(\text{Cond}(E_f))$ is constant across all $\bar{f} \in F_d(\overline{\mathbb{F}_q})$. Thus, the higher direct image $\mathcal{T}_{d, \ell}$ has constant fiber dimension and is locally constant, hence it is a lisse $\mathbb{F}_\ell$ -sheaf on F_d . □

Proposition 2.0.8. Let X be an integral normal scheme with generic point η . Let $j : U \hookrightarrow X$ be the inclusion of a non-empty open subscheme. Let $x \in X \setminus U$ be a point of codimension 1 in X . Write $I_x \subset \pi_1^{\text{ét}}(U, \bar{\eta})$ for the inertia group (Definition 2.0.3).

Let $\mathcal{G}$ be a lisse sheaf on U , in any of the following sense:

1. a Λ -lisse sheaf where Λ is a finite ring,
2. An adic lisse sheaf with coefficients in R , E , or K , where (R, λ) is a complete DVR with R/λ^n finite for all $n \geq 1$, E is its fraction field, and K is an algebraic extension of E ,

Write $\rho : \pi_1^{\text{ét}}(U, \bar{\eta}) \rightarrow \text{GL}(V)$ for the corresponding (Theorem E.0.1) (Theorem E.0.2) continuous representation where V is a module over an appropriate ring.

The geometric stalk $(j_* \mathcal{G})_{\bar{x}}$ is isomorphic to $V^{I_{\bar{x}}}$.

Proof. It suffices to prove this in the case that $\mathcal{G}$ is a lisse sheaf with coefficients in Λ for a finite ring Λ . By ??, we have

$$(j_*\mathcal{G})_{\bar{x}} \cong \Gamma(\mathrm{Spec}(\mathcal{O}_{X,x}^{\mathrm{sh}}), g^*j_*\mathcal{G})$$

where $g : \mathrm{Spec}(\mathcal{O}_{X,x}^{\mathrm{sh}}) \rightarrow X$ is the canonical morphism. Let K_x^{sh} be the fraction field of $\mathcal{O}_{X,x}^{\mathrm{sh}}$. We have the Cartesian square

$$\begin{array}{ccc} K_x^{\mathrm{sh}} & \xrightarrow{\tilde{g}} & U \\ \downarrow \tilde{j} & & \downarrow j \\ \mathrm{Spec}(\mathcal{O}_{X,x}^{\mathrm{sh}}) & \xrightarrow{g} & X. \end{array}$$

By the base change theorem, (♠ TODO: which one)

$$g^*j_*\mathcal{G} \cong \tilde{j}_*\tilde{g}^*\mathcal{G}$$

and hence

$$(j_*\mathcal{G})_{\bar{x}} \cong \Gamma(\mathrm{Spec}(\mathcal{O}_{X,x}^{\mathrm{sh}}), \tilde{j}_*\tilde{g}^*\mathcal{G}) \cong \Gamma(\mathrm{Spec}(K_x^{\mathrm{sh}}), \tilde{g}^*\mathcal{G})$$

(♠ TODO: write as a general fact that inertia group inclusion corresponds to scheme map $\mathrm{Spec}(K_x^{\mathrm{sh}}) \rightarrow U$) The inclusion $\tilde{g} : \mathrm{Spec}(K_x^{\mathrm{sh}}) \rightarrow U$ induces the inclusion $I_x \hookrightarrow \pi_1^{\acute{e}t}(U, \bar{\eta})$ of the inertia group at x . In particular, the pullback $\tilde{g}^*\mathcal{G}$ corresponds to the representation ρ restricted to I_x and its global sections is the I_x -invariants of V . In other words, $(j_*\mathcal{G})_{\bar{x}} \cong V^{I_x}$ as desired. $\square$

Theorem 2.0.9 (Explicit construction of $\mathcal{T}_{d,\ell}$ as a derived pushforward [?, Section 5]). The quotient $\mathcal{G} = \sigma^*\mathcal{E}_{\mathrm{univ}}/\langle(\iota, -1)\rangle$ is the result of ****faithfully flat descent****. The map $\sigma : \tilde{S} \rightarrow S$ is a finite flat Galois cover with group $\mathbb{Z}/2\mathbb{Z}$ over $U = S \setminus U_{\mathrm{ram}}$. The action $(\iota, -1)$ is a ***descent datum***: it specifies that the non-trivial element of $\mathrm{Gal}(\tilde{S}/S) \cong \mathbb{Z}/2\mathbb{Z}$ acts on the elliptic curve fiber via $[-1] \in \mathrm{Aut}(E_1)$. By faithfully flat descent, any scheme over $\tilde{S}$ equipped with a semilinear action of the Galois group descends uniquely to S . The descended object becomes isomorphic to the original over the cover, so it is a ***form*** or ***twist*** of the original. In this case, the descent datum defines an element of

$$H^1(\mathbb{Z}/2\mathbb{Z}, \mathrm{Aut}(E_1)) \cong H^1(K(\sqrt{P})/K, \mathrm{Aut}(E_1)),$$

which classifies quadratic twists of E_1 . More generally, twists exist for any finite Galois extension L'/K , and are classified by $H^1(\mathrm{Gal}(L'/K), \mathrm{Aut}(E))$; quadratic twists correspond to the degree-2 case.

This is the quadratic twist of $\mathcal{E}_{\mathrm{univ}}$ by $P(t, c)$, given in the model produced by Galois descent. Restricting to any fiber $C \times \{\bar{f}\}$ where $c \mapsto f$, this gives the quadratic twist of $\mathcal{E}_1$ by $f \in K^\times/(K^\times)^2$.

Thus the fiber of $\mathcal{G}$ over $(t, c) \in S$ is the quadratic twist of the fiber of $\mathcal{E}_1$ at t by the value $P(t, c) \in K^\times/(K^\times)^2$. In particular, for each fixed $\bar{f} \in F_d(\overline{\mathbb{F}}_q)$ with corresponding polynomial f , the restriction of $\mathcal{G}$ to $C \times \{\bar{f}\}$ is canonically isomorphic to $\mathcal{E}_f \rightarrow C$.

Let $\mathcal{G}_\ell \subset \mathcal{G}$ be the kernel of multiplication by ℓ . This is a finite étale group scheme over S . Let $j : S_{\text{sm}} \hookrightarrow S$ be the open immersion of the locus where $\mathcal{G} \rightarrow S$ has good or multiplicative reduction (the complement of the additive reduction locus). The middle extension

$$j_* j^* \mathcal{G}_\ell$$

is a constructible $\mathbb{F}_\ell$ -sheaf on S . By assumption on ℓ , $\mathcal{G}_\ell$ is already a middle extension over each fiber $C \times \{\bar{f}\}$.

Let $\pi : S = C \times F_d \rightarrow F_d$ be the projection. The first higher direct image

$$\mathcal{T}_{d,\ell} := R^1 \pi_* (j_* j^* \mathcal{G}_\ell)$$

is a lisse $\mathbb{F}_\ell$ -sheaf on F_d .

For each geometric point $\bar{f} \in F_d(\bar{\mathbb{F}}_q)$, the geometric fiber is

$$\mathcal{T}_{d,\ell}|_{\bar{f}} \cong H^1(C \times \bar{\mathbb{F}}_q, \mathcal{E}_{f,\ell}),$$

where $\mathcal{E}_{f,\ell}$ is the Néron model of the ℓ -torsion $E_f[\ell]$ for the quadratic twist E_f/K .

Remark 2.0.10 (Orthogonal pairing). The Weil pairing on $E_f[\ell] \times E_f[\ell] \rightarrow \mu_\ell$ extends to a non-degenerate alternating pairing $\mathcal{E}_{f,\ell} \times \mathcal{E}_{f,\ell} \rightarrow \mathbb{F}_\ell(1)$, making $\mathcal{E}_{f,\ell}$ self-dual. Together with Poincaré duality this induces a non-degenerate symmetric (orthogonal) pairing on the cohomology

$$H^1(C \times \bar{\mathbb{F}}_q, \mathcal{E}_{f,\ell}) \times H^1(C \times \bar{\mathbb{F}}_q, \mathcal{E}_{f,\ell}) \longrightarrow \mathbb{F}_\ell.$$

The sheaf $\mathcal{T}_{d,\ell}$ carries this orthogonal pairing fiberwise, so the monodromy group of $\mathcal{T}_{d,\ell}$ lies in the orthogonal group $O(V_{f,\ell})$.

3. ELLENBERG AND LANDESMAN'S NEEDS

Definition 3.0.1 (see [?, Definition 2.4.2.] for a discussion). (♠ TODO: divisor over a scheme) Let B be a base scheme. Let $C \rightarrow B$ be a smooth, proper relative curve with geometrically connected fibers of genus g . Let $Z \subset C$ be a finite étale divisor over B . Let $U = C \setminus Z$.

(♠ TODO: boundary component, puncture) Let $\Sigma_{g,n,f+1}$ a genus g topological surface with b boundary components and f punctures. (♠ TODO: action) Suppose that $\mathcal{S} \subset \text{Hom}(\pi_1^{\text{ét}}(\Sigma_{g,n,f+1}), G)$ is a G conjugation invariant subset preserved by the action of $\pi_1^{\text{ét}}(\text{Conf}_{U/\bar{b}}^n)$ acting on the first n points. Define the **Hurwitz stack** $\text{Hur}_{C/B}^{G,n,Z,S}$ to be the algebraic stack over B whose functor of points is defined as follows: for T a B -scheme, $\text{Hur}_{C/B}^{G,n,Z,S}(T)$ is the groupoid whose objects are the tuples

$$(D, i : D \rightarrow C_T, X, h : X \rightarrow C_T)$$

such that

- D is a finite étale cover of T of degree n .
- i is a closed immersion $i : D \subset C_T$ that is disjoint from $Z_T \subset C_T$.
- X is a smooth proper relative curve over T , not necessarily having geometrically connected fibers.

- $h : X \rightarrow C_T$ is a finite locally free Galois G -cover that is étale away from $Z_T \cup i(D) \subset C_T$.
- Let $\bar{t} \rightarrow T$ be a fixed geometric point. Let $\bar{\eta}$ denote the geometric generic point of $(C_T)_{\bar{t}}$. Then the representation $\rho : \pi_1((U_T)_{\bar{t}} - i(D_{\bar{t}}), \bar{\eta}) \rightarrow G$ corresponding to the G -cover h , under the identification of $\text{Hom}(\pi_1((U_T)_{\bar{t}} - i(D_{\bar{t}}), \bar{\eta}), G)$ and $\text{Hom}(\pi_1(\Sigma_{g,n+f+1}), G)$ corresponds to an element of $\mathcal{S}$.

Two covers $X \rightarrow C_T$ are considered equivalent if they are related by the G -conjugation action (♠ TODO: what does this mean). Morphisms between two points (D_j, i_j, X_j, h_j) for $j \in \{1, 2\}$ are given by (ϕ_D, ψ_X) where $\phi_D : D_1 \simeq D_2$ is an isomorphism so that $i_2 \circ \phi_D = i_1$ and $\psi_X : X_1 \simeq X_2$ is an isomorphism such that $h_2 \circ \psi_X = h_1$ and $\psi_X = g^{-1} \psi_X g$ for every $g \in G$.

Definition 3.0.2. Let B be a base scheme. Let $C \rightarrow B$ be a smooth, proper relative curve with geometrically connected fibers of genus g . Let $Z \subset C$ be a finite étale divisor over B . Let $U = C \setminus Z$.

Proposition 3.0.3. Let $B = \bar{b} = \text{Spec } k$ be a geometric point where k is an algebraically closed field.

(♠ TODO: say what g,r,f, Dx, F) are Let n be an even integer such that

$$n > \max \left(2g, \frac{2(2r+1)(f+1) - \sum_{y \in D_x(\bar{b})} \text{Drop}_y(\mathcal{F})}{2r} - (2g-2) \right).$$

(Definition E.0.6) Let $C \rightarrow B$ be a smooth, proper connected curve of genus g . Let $Z \subset C$ be a finite étale divisor over B . Let $U = C \setminus Z$. Let $\text{QTwist}_{U/B}^n := \text{Hur}_{C/B}^{\mathbb{Z}/2\mathbb{Z}, n, Z, \mathcal{S}}$ (Definition 3.0.1); note that there is a natural projection map

$$\text{QTwist}_{U/B}^n \rightarrow \text{Conf}_{U/B}^n.$$

(♠ TODO: this variant of the configuration space) Let $x = [(D, \phi)] \in \text{QTwist}_{U/B}^n$ be a k -point, let $y \in \text{Conf}_{U/B}^n$ be its image under the above projection map. Let

$$\mathcal{C}_B^n := \text{Conf}_{U/B}^n \times_B C$$

be the “universal curve” over the configuration space $\text{Conf}_{U/B}^n$ associated to the curve $C \rightarrow B$, and let

$$\xi : \mathcal{C}_B^n \rightarrow \text{Conf}_{U/B}^n$$

be the projection map. Let $C_x \cong y \times_B C \cong C$ be the fiber of ξ over y .

By (♠ TODO: cite katz), there is a map $h : C_x \rightarrow \mathbb{P}^1$ of degree n such that the branch locus $\text{br}(h)$ of h is disjoint from $h((Z \cup D)_x)$, h separates points of $(Z \cup D)_x$, and precisely one point $\delta \in D_x$ maps to $\infty \in \mathbb{P}^1$. Let $W \subset \mathbb{P}^1$ to be the complement of $\text{br}(h) \cup h(Z \cup D)$.

(♠ TODO: wait, how do we define kummer sheaves over more general fields?) We construct a map $\phi : W \rightarrow \text{QTwist}_{U/\bar{b}}^n$. Such a map corresponds to a finite double cover V of $C \times_{\bar{b}} W$, which in turn is equivalent to specifying a rank 1 locally constant constructible $\mathbb{Z}/\ell\mathbb{Z}$ -sheaf on an open subset of $C \times_{\bar{b}} W$ whose monodromy is trivialized by. Let $t : X \rightarrow U$ be the finite double cover corresponding to x ; in particular, t is étale away from D . Let $\mathbb{V} := t_*(\mathbb{Z}/\ell\mathbb{Z})/(\mathbb{Z}/\ell\mathbb{Z})$,

which is a rank 1 constructible sheaf $\mathbb{Z}/\ell\mathbb{Z}$ -sheaf on U that becomes locally constant on $U - D$. Let $\mathcal{F}' = \mathcal{F} \otimes \mathbb{V}$, i.e. $\mathcal{F}'$ is the quadratic twist of $\mathcal{F}$ corresponding to x . (♠ TODO: think about how $\mathcal{F}'$ is the quadratic twist corresponding to x)

Let $\mathcal{L}_\chi$ be the Kummer sheaf (Definition 2.0.2) on $\mathbb{G}_m$ corresponding to the double cover $\mathbb{G}_m \xrightarrow{z \mapsto z^2} \mathbb{G}_m$. Construct the following maps:

$$\begin{aligned} \alpha' : \mathbb{A}^1 \times \mathbb{A}^1 - \Delta &\rightarrow \mathbb{G}_m, \quad (x, y) \mapsto x - y \\ \alpha : Y &:= (h, \text{id})^{-1}(W \times W - \Delta) \xrightarrow{(h, \text{id})} \mathbb{A}^1 \times \mathbb{A}^1 - \Delta \xrightarrow{\alpha'} \mathbb{G}_m. \end{aligned}$$

let $\mathbb{W} := \alpha^* \mathcal{L}_\chi$. Let $p_C : C \times W \rightarrow C$ be the projection. (♠ TODO: continue) Noting that $h^{-1}(W - 0) \subset U - D$, let $\mathbb{V}'$ be the sheaf on $h^{-1}(W - 0)$ given by

$$\mathbb{V}' := \mathbb{W}|_{h^{-1}(W-0) \times 0} \otimes \mathbb{V}|_{h^{-1}(W-0)}.$$

4. TWISTS

Definition 4.0.1. (♠ TODO: AI generated, verify) Let U be a connected scheme, and let $\bar{u} : \text{Spec}(\Omega) \rightarrow U$ be a geometric point. Let $\pi_1^{\text{ét}}(U, \bar{u})$ denote the étale fundamental group of U at $\bar{u}$.

Let Λ be one of the following coefficient rings:

1. A finite ring,
2. A complete DVR R ,
3. The field of fractions E of a complete DVR R , or
4. An algebraic extension of the field of fractions E of a complete DVR R .

If Λ is a complete DVR, a fraction field thereof, or an algebraic extension of such a fraction field, then further assume that U is a connected noetherian scheme.

1. Let A be one of the coefficient rings listed above. A *character on $\pi_1^{\text{ét}}(U, \bar{u})$ with values in A* is a continuous group homomorphism

$$\chi : \pi_1^{\text{ét}}(U, \bar{u}) \rightarrow A^\times$$

where $A^\times$ denotes the group of units of A , equipped with the discrete topology.

2. Let $\chi : \pi_1^{\text{ét}}(U, \bar{u}) \rightarrow A^\times$ be a character as in (1). By the equivalence of categories between lisse sheaves on $U_{\text{ét}}$ and continuous representations of $\pi_1^{\text{ét}}(U, \bar{u})$ (Theorem E.0.1) (Theorem E.0.2), the character χ corresponds uniquely to a rank-1 lisse sheaf, denoted $\mathcal{L}_\chi$, often called the *rank-1 lisse sheaf associated to χ* . The stalks of $\mathcal{L}_\chi$ are isomorphic to A as A -modules, and the action of $\gamma \in \pi_1^{\text{ét}}(U, \bar{u})$ on a stalk is multiplication by $\chi(\gamma)$.
3. Let $\mathcal{F}$ be a lisse sheaf with coefficients in A , corresponding to a representation $\rho : \pi_1^{\text{ét}}(U, \bar{u}) \rightarrow \text{Aut}_A(V)$, and let χ be a character as above. The *twist of $\mathcal{F}$ by χ* , denoted $\mathcal{F}_\chi$, is the lisse sheaf $\mathcal{F} \otimes \mathcal{L}_\chi$ corresponding to the tensor product representation

$$(\rho \otimes \chi)(\gamma) = \chi(\gamma) \cdot \rho(\gamma)$$

for all $\gamma \in \pi_1^{\text{ét}}(U, \bar{u})$. Since $\mathcal{L}_\chi$ has rank 1, the stalks of $\mathcal{F} \otimes \mathcal{L}_\chi$ are isomorphic to those of $\mathcal{F}$ as A -modules, but the monodromy action is scaled by the character χ .

4. Two lisse sheaves $\mathcal{F}$ and $\mathcal{G}$ with coefficients in A are called *twists of each other* if there exists a character $\chi : \pi_1^{\text{ét}}(U, \bar{u}) \rightarrow A^\times$ such that

$$\mathcal{G} \cong \mathcal{F} \otimes \mathcal{L}_\chi.$$

APPENDIX A. ABELIAN VARIETIES

Definition A.0.1. Let k be a field. A *(algebraic) variety over k* is an integral, separated scheme of finite type over k .

Definition A.0.2. Let S be a scheme. An *algebraic group scheme over S* (or an *S -group scheme*) is a group object G in the category of schemes over S ; that is, G is an S -scheme equipped with S -morphisms: $m : G \times_S G \rightarrow G$ (*multiplication*), $i : G \rightarrow G$ (*inverse*), and $e : S \rightarrow G$ (*identity/identity section*), satisfying the group axioms expressed by the commutativity of the following diagrams:

$$\begin{array}{lcl}
\text{1. Associativity} & \begin{array}{ccc} G \times_S G \times_S G & \xrightarrow{m \times \text{id}} & G \times_S G \\ \text{id} \times m \downarrow & & \downarrow m \\ G \times_S G & \xrightarrow{m} & G \end{array} & \\
\text{2. Identity} & \begin{array}{ccc} G \times_S S & \xrightarrow{\text{id} \times e} & G \times_S G \\ \searrow \cong & \downarrow m & \\ & G & \end{array} & \begin{array}{ccc} S \times_S G & \xrightarrow{e \times \text{id}} & G \times_S G \\ \searrow \cong & \downarrow m & \\ & G & \end{array} \\
\text{3. Inverse} & \begin{array}{ccc} G & \xrightarrow{(\text{id}, i)} & G \times_S G \\ \text{id} \downarrow & & \downarrow m \\ G & \xrightarrow{e \circ \pi} & G \end{array} & \text{where } \pi : G \rightarrow S \text{ is the structure morphism and } e \circ \pi \\
& & & \text{sends } g \text{ to the identity section.}
\end{array}$$

If the multiplication map also satisfies the commutative axiom, then the group scheme G is called *commutative* or *abelian*.

If G is affine over S , we call it an *affine group scheme over S* .

If the base scheme S is the spectrum of a field k , then we call G a *k -algebraic group* or an *algebraic group (scheme) over k* . If G is additionally a k -variety, then we call G a *k -group variety*.

Definition A.0.3. Let S be a scheme, and let G and H be S -algebraic groups (Definition A.0.2). A morphism of S -schemes $f : G \rightarrow H$ is a *homomorphism of algebraic groups* if f is a group homomorphism, i.e.,

- $f(m_G(x, y)) = m_H(f(x), f(y))$ for all $x, y \in G$,
- $f(i_G(x)) = i_H(f(x))$ for all $x \in G$, and

- $f(e_G) = e_H$.

It is called an *isomorphism of algebraic groups (over S)* if it is additionally an isomorphism of S -schemes. If there exists an isomorphism $f : G \rightarrow H$ of algebraic groups over S , then G and H are said to be *isomorphic S -algebraic groups*.

Definition A.0.4 (Kernel of a homomorphism of group schemes over a scheme). Let S be a scheme and let $\phi : G_1 \rightarrow G_2$ be a homomorphism of algebraic group schemes over S (Definition A.0.3).

The *kernel of ϕ* , denoted by $\ker \phi$ or $A[\phi]$, is the fiber of ϕ at the identity section (Definition A.0.2) $e_2 : S \rightarrow G_2$ of G_2 . Equivalently, $\ker \phi$ is the subgroup scheme of G_1 defined as the pullback:

$$\ker \phi = G_1 \times_{G_2} S,$$

(♠ TODO: finite group scheme) where the map $G_1 \rightarrow G_2$ is ϕ and the map $S \rightarrow G_2$ is the identity section. If ϕ is a finite morphism, then $\ker \phi$ is a finite group scheme over S .

(♠ TODO: group variety)

Definition A.0.5 (Abelian variety over a field). Let k be a field. An *abelian variety over k* is a complete, connected algebraic group variety defined over k , i.e.

- A is a smooth, proper, geometrically connected algebraic variety over k ,
- A is endowed with a group structure defined by morphisms of varieties over k (multiplication $m : A \times_k A \rightarrow A$ and inverse $i : A \rightarrow A$),
- the group law satisfies the group axioms scheme-theoretically.

In particular, an abelian variety is a projective algebraic group variety (Definition A.0.2) over k .

Definition A.0.6 (Isogeny of abelian varieties over a field). (♠ TODO: finite group scheme) Let k be a field and let A and B be abelian varieties (Definition A.0.5) over k . An *isogeny of abelian varieties over k* is a homomorphism (Definition A.0.3) $\varphi : A \rightarrow B$ of group schemes over k that is surjective and has finite kernel (Definition A.0.4). The kernel $\ker \varphi$ is a finite group scheme over k (Definition A.0.4). Equivalently, φ is a finite, faithfully flat homomorphism of abelian varieties over k .

A.1. Picard groups and Jacobians.

Definition A.1.1. (♠ TODO: picard group of an algebraic space) Let X be a scheme. The *Picard group of X* , denoted by $\text{Pic}(X)$, is the group of isomorphism classes of invertible sheaves (line bundles) on X (with respect to the Zariski topology), with the group operation induced by the tensor product of sheaves.

More precisely:

1. Two invertible sheaves $\mathcal{L}$ and $\mathcal{M}$ are isomorphic if there exists an $\mathcal{O}_X$ -module isomorphism $\phi : \mathcal{L} \xrightarrow{\sim} \mathcal{M}$.

2. The tensor product of invertible sheaves induces a well-defined group operation on the set of isomorphism classes. (♠ TODO: sheafy dual)
3. The inverse of an invertible sheaf $\mathcal{L}$ is given by its dual $\mathcal{L}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$.
4. The identity element is the class of the structure sheaf $\mathcal{O}_X$.

(♠ TODO: class group of ring) When $X = \text{Spec}(A)$ is an affine scheme, there is a canonical isomorphism

$$\text{Pic}(\text{Spec}(A)) \cong \text{Cl}(A),$$

where $\text{Cl}(A)$ is the class group of the ring A (the group of isomorphism classes of rank-1 projective modules over A , also known as the Picard group of the ring).

(♠ TODO: extract this out) If X is a Noetherian, integral, and locally factorial scheme, then $\text{Pic}(X)$ is isomorphic to the Weil divisor class group $\text{Cl}(X)$.

Definition A.1.2. Let X/k be a smooth, geometrically integral projective variety over a field k .

The relative Picard functor $\text{Pic}_{X/k}$ is representable by a smooth scheme over k . (♠ TODO: show that the identity component is an abelian variety) The identity component $\text{Pic}_{X/k}^0$ is in fact an abelian variety and is called the *Picard variety of X/k* .

Definition A.1.3 (Jacobian variety of a curve). (♠ TODO: separate out jacobian of just a proper curve and the jacobian of a nice curve) Let X be a proper curve over a field k .

1. The relative Picard functor $\text{Pic}_{X/k}$ is representable by a smooth group scheme (Definition A.0.2) over k . The *(generalized) Jacobian of X* refers to the identity component $\text{Pic}_{X/k}^0$ of this group scheme.
2. Now assume that X is smooth, is projective, and has geometrically connected components. The Jacobian is an abelian variety (Definition A.0.5) over k and may be called the *Jacobian variety of X* . It may be denoted by notations such as $\text{Jac}(X)$, Jac_X , or J_X .

Under these assumptions, note that $\text{Jac}(X)$ coincides with the Picard variety (Definition A.1.2) of X/k . Equivalently, J_X parametrizes isomorphism classes of degree-zero line bundles on X , where a line bundle $\mathcal{L}$ has *degree zero* if its degree (computed as the sum of orders of any rational function defining it) vanishes.

The Jacobian variety $\text{Jac}(X)$ has dimension g where g is the geometric genus of X .

When X is equipped with a base point, the Jacobian variety carries a canonical principal polarization Θ , making (J_X, Θ) a principally polarized abelian variety.

APPENDIX B. NÉRON MODELS OF ABELIAN VARIETIES

Definition B.0.1 (Dedekind domain). An integral domain R is called a *Dedekind domain* or a *Dedekind ring* if it satisfies the following equivalent conditions:

- R is Noetherian, integrally closed in its field of fractions, and every nonzero prime ideal of R is maximal.

- Equivalently: for every nonzero prime ideal $\mathfrak{p}$ of R , the localization $R_{\mathfrak{p}}$ is a discrete valuation ring.

Definition B.0.2 (Dedekind scheme). A *Dedekind scheme* is a scheme S that is integral, Noetherian, normal, and of Krull dimension 1. Equivalently, every point $s \in S$ has an open neighborhood $U = \text{Spec } R$ such that R is a Dedekind domain (Definition B.0.1). In particular, for every point $s \in S$, the local ring $\mathcal{O}_{S,s}$ is either a field (if s is the generic point) or a discrete valuation ring (if s is a closed point).

Definition B.0.3 (Special fiber and generic fiber). Let S be an integral scheme, let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S . Let $s \in S$ be a closed point.

1. For any morphism of schemes $f : X \rightarrow S$, the *generic fiber of X over S* (or simply the *generic fiber*) is the base change:

$$X_{\eta} = X \times_S \text{Spec } K$$

of f along $\text{Spec } K \rightarrow S$.

2. For any morphism of schemes $f : X \rightarrow S$, the *special fiber of X at s* is the base change:

$$X_s = X \times_S \text{Spec } k(s)$$

of f along $\text{Spec } k(s) \rightarrow S$, where $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$ is the residue field at s .

In the case that S has only closed point (e.g. S is the ring of integers of a local field), the *special fiber of X* refers to the special fiber of X at this unique closed point.

Definition B.0.4 (Néron model of an abelian variety). Let S be an integral scheme, let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S . Let A_K be an abelian variety over K (Definition A.0.5).

A *Néron model of A_K over S* is a smooth separated group scheme $\mathcal{A}$ of finite type over S whose generic fiber (Definition B.0.3) $\mathcal{A}_{\eta}$ is isomorphic to A_K , satisfying the following universal property (the *Néron mapping property*): for every smooth S -scheme U and every K -morphism $f_K : U_{\eta} \rightarrow A_K$, there exists a unique S -morphism $f : U \rightarrow \mathcal{A}$ extending f_K .

If such a model exists, it is unique up to unique isomorphism.

APPENDIX C. GALOIS COHOMOLOGY

Definition C.0.1. Let G be a group and R a not necessarily commutative ring. Let A be a left/right $R[G]$ -module.

1. The *cohomology group of G with coefficients in A* or the *group cohomology of A as an $R[G]$ -module*, denoted $H^n(G, A)$ or $H^n(G; A)$ for $n \geq 0$, is the n -th right derived functor object $(R^n(-^G))(A)$ of the invariants functor $A \mapsto A^G$, which is left exact by ??.
2. The *homology group of G with coefficients in A* or the *group homology of A as an $R[G]$ -module*, denoted $H_n(G, A)$ or $H_n(G; A)$, for $n \geq 0$, is the n -th left derived functor object $(L_n(-_G))(A)$ of the coinvariants functor $A \mapsto A_G$, which is right exact by ??.

Definition C.0.2. Let R be a topological ring and G a topological group. Let $R\text{-}\mathbf{TopMod}_G$ be the category of (left) topological G -modules over R .

Let M be an object in $R\text{-}\mathbf{TopMod}_G$. The *continuous cochain complex* $C_{\text{cont}}^\bullet(G, M)$ (also denoted by $C^\bullet(G, M)$, but which should not be confused with the cochain complex $C^\bullet(G, M)$ that does not incorporate any topological structure) is defined as follows:

1. $C_{\text{cont}}^n(G, M) = \text{Map}_{\text{cont}}(G^n, M)$, the R -module of continuous maps from G^n to M .
2. The *coboundary map* $d^n : C_{\text{cont}}^n(G, M) \rightarrow C_{\text{cont}}^{n+1}(G, M)$ is given by:

$$(d^n f)(g_1, \dots, g_{n+1}) = g_1 f(g_2, \dots, g_{n+1}) + \sum_{i=1}^n (-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) + (-1)^{n+1} f(g_1, \dots, g_n)$$

Definition C.0.3. Let R be a topological ring and G a topological group. Let $R\text{-}\mathbf{TopMod}_G$ be the category of (left) topological G -modules over R .

The *continuous cohomology groups* $H_{\text{cont}}^n(G, M)$ are defined as the cohomology groups of the continuous cochain complex (Definition C.0.2) $C_{\text{cont}}^\bullet(G, M)$.

The continuous cohomology group is also denoted by $H^n(G, M)$ if the continuity is clear from context, but this should not be confused for the (non-continuous) cohomology group (Definition C.0.1) $H^n(G, M)$ of M as a G -module (without consideration of the topological structures).

Theorem C.0.4. (♠ TODO: verify these facts)

Let R be a topological ring and G a topological group. Let $R\text{-}\mathbf{TopMod}_G$ be the category of (left) topological G -modules over R .

Consider the G -invariant functor $(-)^G : R\text{-}\mathbf{TopMod}_G \rightarrow R\text{-}\mathbf{TopMod}$ (??).

Assume either of the following:

1. G is profinite, R is discrete, and M is discrete; in this case, by the “right derived functors of $(-)^G$ ”, we mean the right derived functor on the category of discrete G -modules, which has enough injectives (♠ TODO: cite; apparently the category of discrete G -modules over a topological ring R where G is profinite is an abelian category with enough injectives).
2. G is locally compact, and M is complete; in this case, by the “right derived functors of $(-)^G$ ”, we mean the right derived functor in the sense of relative homological algebra, using resolutions that split as sequences of topological spaces (♠ TODO: figure out what this means).

The continuous cohomology groups (Definition C.0.3) $H_{\text{cont}}^n(G, M)$, i.e. the cohomology of the continuous cochain complex (Definition C.0.2) $C_{\text{cont}}^\bullet(G, M)$, is naturally isomorphic to the the right derived functors of $(-)^G$. In other words,

$$H^n(C_{\text{cont}}^\bullet(G, M)) \cong (R^n(-^G))(M)$$

Definition C.0.5. Let R be a topological ring (e.g., a ℓ -adic ring $\mathbb{Z}_\ell$).

1. Let L/K be a Galois field extension. Let M be a Galois module of $\text{Gal}(L/K)$ over R . The *Galois cohomology group of L/K with coefficients in M* is:

$$H^n(L/K, M) = H_{\text{cont}}^n(\text{Gal}(L/K), M)$$

(Definition C.0.3).

2. Let K be a field with absolute Galois group $G_K = \text{Gal}(\bar{K}/K)$. Let M be a topological G -module over R .

The *Galois cohomology group of K with coefficients in M* is:

$$H^n(K, M) = H^n(\bar{K}/K, M) = H_{\text{cont}}^n(\text{Gal}(\bar{K}/K), M).$$

In either case, one also lets $H^n(G, M)$ denote the Galois cohomology groups where G is the relevant Galois group.

APPENDIX D. SELMER GROUPS

Definition D.0.1 (Kummer map associated to an isogeny of an abelian variety). Let k be a field, and let A and B be commutative group schemes (Definition A.0.2) over S . Suppose $\varphi : A \rightarrow B$ is an isogeny over S . Denote by $G_k = \text{Gal}(\bar{k}/k)$ the absolute Galois group of k .

The short exact sequence

$$0 \rightarrow \ker \varphi \rightarrow A(\bar{k}) \xrightarrow{\varphi} B(\bar{k}) \rightarrow 0.$$

yields the following long exact sequence in Galois cohomology (Definition C.0.5):

$$\begin{aligned} 0 \rightarrow \ker \varphi(k) \rightarrow A(k) \xrightarrow{\varphi} B(k) \\ \xrightarrow{\delta_\varphi} H^1(G_k, \ker \varphi) \rightarrow H^1(G_k, A(\bar{k})) \rightarrow H^1(G_k, B(\bar{k})) \rightarrow \cdots \end{aligned}$$

The *Kummer map associated to the isogeny φ* is the Galois cohomological connecting homomorphism

$$\delta_\varphi : B(k) \rightarrow H^1(G_k, \ker \varphi),$$

above. In fact, there is a short exact sequence

$$0 \rightarrow B(k)/\varphi(A(k)) \xrightarrow{\delta'_\varphi} H^1(G_k, \ker \varphi) \rightarrow H^1(G_k, A)[\varphi] \rightarrow 0$$

where the map δ'_φ is induced by δ_φ . We may also let the *Kummer map associated to φ* refer to this map δ'_φ . We may also use the abuse of notation δ_φ to denote δ'_φ .

Definition D.0.2 (Selmer group for an isogeny of abelian group schemes over a global field). (♠ TODO: It should be possible to define this for more general group schemes) Let K be a global field, and let A and B be abelian group schemes (Definition A.0.2) defined over K . Suppose $\varphi : A \rightarrow B$ is an isogeny defined over K .

Denote by $G_K = \text{Gal}(\bar{K}/K)$ the absolute Galois group of K . For each place v of K , fix an extension v to $\bar{K}$, which yields an embedding $\bar{K} \subset \bar{K}_v$ and a decomposition group (Definition E.0.3) $G_v \subset G_K$.

The *Selmer group of φ over K* , denoted by notations such as $\text{Sel}(A/K)$, $\text{Sel}^\varphi(A/K)$, $\text{Sel}^{(\varphi)}(A/K)$, $\text{Sel}_\varphi(A/K)$, is the subgroup of the Galois cohomology group (Definition C.0.5) $H^1(K, \ker \varphi)$ given by

$$\text{Sel}^\varphi(A/K) := \ker \left(H^1(G_K, A[\varphi]) \rightarrow \prod_v H^1(G_v, A(\overline{K}))[\varphi] \right),$$

(♠ TODO: define the kummer map associated to φ) where the product runs over all places v of K , and $H^1(G_v, A)[\varphi]$ denotes the image of the local Kummer map associated to φ .

We describe the map

$$(A) \quad H^1(G_K, A[\varphi]) \rightarrow \prod_v H^1(G_v, A(\overline{K}))[\varphi]$$

(♠ TODO: define a decomposition group) used to define the kernel above: Note that G_v acts on $A(\overline{K}_v)$ and $B(\overline{K}_v)$, and the base change φ_v of φ to K_v induces a Kummer short exact sequence (Definition D.0.1)

$$0 \rightarrow B(K_v)/\varphi(A(K_v)) \xrightarrow{\delta_{\varphi_v}} H^1(G_v, A[\varphi]) \rightarrow H^1(G_v, A(\overline{K}_v))[\varphi] \rightarrow 0.$$

The natural inclusions $G_v \subset G_K$ and $E(K) \subset E(K_v)$ give restriction maps $H^1(G_K, A[\varphi]) \rightarrow H^1(G_v, A[\varphi])$ on Galois cohomology, and we have a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & B(K)/\varphi(A(K)) & \xrightarrow{\delta_\varphi} & H^1(G_K, A[\varphi]) & \longrightarrow & H^1(G_K, A(\overline{K}))[\varphi] \longrightarrow 0 \\ & & \downarrow & & \downarrow \Pi_v \text{ res}_v & \searrow \text{dotted} & \downarrow \\ 0 & \longrightarrow & \prod_v B(K_v)/\varphi(A(K_v)) & \xrightarrow{\delta_{\varphi_v}} & \prod_v H^1(G_v, A[\varphi]) & \longrightarrow & \prod_v H^1(G_v, A(\overline{K}_v))[\varphi] \longrightarrow 0 \end{array}$$

The map (A) is the one given in the above commutative diagram by the dotted arrow.

Equivalently,

$$\text{Sel}^\varphi(A/K) = \{c \in H^1(K, A[\varphi]) : \text{for all places } v \text{ of } K, \text{res}_v(c) \in \text{Im } \delta_{\varphi_v}\}$$

APPENDIX E. ÉTALE SHEAVES

Theorem E.0.1 (See [?, Tags 0DV5, 0GIY]). Let X be a connected scheme and let $\bar{x} \in X$ be a geometric point.

1. There is an equivalence of categories (♠ TODO: define finite in this context)

$$\left\{ \begin{array}{l} \text{finite locally constant} \\ \text{sheaves of sets on } X_{\text{étale}} \end{array} \right\} \longleftrightarrow \{ \text{finite } \pi_1^{\text{ét}}(X, \bar{x})\text{-sets} \}.$$

2. There is an equivalence of categories

$$\left\{ \begin{array}{l} \text{finite cardinality locally constant} \\ \text{sheaves of abelian groups on } X_{\text{étale}} \end{array} \right\} \longleftrightarrow \{ \text{finite cardinality } \pi_1^{\text{ét}}(X, \bar{x})\text{-modules} \}.$$

3. For a finite ring Λ , there is an equivalence of categories

$$\left\{ \begin{array}{l} \text{finite type, locally constant} \\ \text{sheaves of } \Lambda\text{-modules on } X_{\text{étale}} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{finite } \pi_1^{\text{ét}}(X, \bar{x})\text{-modules endowed} \\ \text{with commuting } \Lambda\text{-module structure} \end{array} \right\}.$$

()

(♠ TODO: in the below, the geometrically unibranch hypothesis might not be needed, but it was in the stacks project)

4. Assume that X is irreducible and geometrically unibranch. For a ring Λ , there is an equivalence of categories

$$\left\{ \begin{array}{l} \text{finite type, locally constant} \\ \text{sheaves of } \Lambda\text{-modules on } X_{\text{étale}} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{finite type } \Lambda\text{-modules } M \text{ endowed} \\ \text{with a continuous } \pi_1^{\text{ét}}(X, \bar{x})\text{-action} \end{array} \right\}.$$

See also Theorem E.0.2.

Theorem E.0.2. Let (R, λ) be a complete DVR such that R/λ^n is finite for all $n \geq 1$. Let E be the fraction field of R . Let K be an algebraic extension of E . Let X be a connected noetherian scheme. Let $\bar{x} \in X$ be a geometric point.

1. There is an equivalence of categories between:

$$\{\text{lisse } \lambda\text{-adic sheaves on } X_{\text{étale}}\} \longleftrightarrow \left\{ \begin{array}{l} \text{finitely generated free } R\text{-modules } M \\ \text{with a continuous } \pi_1^{\text{ét}}(X, \bar{x})\text{-action} \end{array} \right\}.$$

2. There is an equivalence of categories between:

$$\{\text{lisse objects of } \text{Sh}(X, E)\} \longleftrightarrow \left\{ \begin{array}{l} \text{finite-dimensional } E\text{-vector spaces } V \\ \text{with a continuous } \pi_1^{\text{ét}}(X, \bar{x})\text{-action} \end{array} \right\}.$$

3. Let K be an algebraic extension of E . There is an equivalence of categories between:

$$\{\text{lisse objects of } \text{Sh}(X, K)\} \longleftrightarrow \left\{ \begin{array}{l} \text{finite-dimensional } K\text{-vector spaces } V \\ \text{with a continuous } \pi_1^{\text{ét}}(X, \bar{x})\text{-action} \end{array} \right\}.$$

Definition E.0.3. Assume ?? and further assume that L/K is a Galois extension.

The *decomposition group of L/K* is the subgroup

$$D_w = \{\sigma \in \text{Gal}(L/K) : w \circ \sigma = w\}.$$

It is also the inverse limit of the decomposition groups of the finite Galois subextensions E/K within L (with valuation restricted from the valuation of L). It is a closed subgroup of $\text{Gal}(L/K)$.

Letting (K, v) be a valuation field, its *decomposition group* $D_{K,v}$ (also sometimes denoted by $G_{K,v}$) is the decomposition group of K^{sep}/K in the above sense where a valuation on K^{sep} extending v is fixed.

Definition E.0.4. Assume ?? and further assume that L/K is a Galois extension.

There is a natural surjective homomorphism

$$D_w \rightarrow \text{Gal}(k_w/k_v)$$

(Definition E.0.3) on the residue fields. The *inertia group of L/K* is its kernel:

$$I_w = \{\sigma \in D_w : \sigma(x) \equiv x \pmod{\mathfrak{m}_v} \text{ for all } x \in \mathcal{O}_v\}.$$

(Definition E.0.3) It is also the inverse limit of the decomposition groups of the finite Galois subextensions E/K within L (with valuation restricted from the valuation of L). It is a closed subgroup of D_w .

Letting (K, v) be a valuation field, its *inertia group $I_{K,v}$* is the inertia group of K^{sep}/K in the above sense where a valuation on K^{sep} extending v is fixed.

Proposition E.0.5 (see [?, Tag 0BQM]). Let X be a normal integral scheme function field K . Let η be the generic point of X and let $\bar{\eta}$ be a geometric point of X whose image is η . The canonical continuous group homomorphism

$$\text{Gal}(K^{\text{sep}}/K) \cong \pi_1^{\text{ét}}(\eta, \bar{\eta}) \rightarrow \pi_1(X, \bar{\eta})$$

(♠ TODO: better describe the canonical map)

is identified with the quotient map

$$\text{Gal}(K^{\text{sep}}/K) \rightarrow \text{Gal}(M/K)$$

(♠ TODO: defined unramifiedness of X in L) where $M \subseteq K^{\text{sep}}$ is the union of the finite subextensions L such that X is unramified in L .

Definition E.0.6. (♠ TODO: AI generated, verify) Let X be an integral normal scheme with generic point η . Let $\bar{\eta}$ be a geometric point of X whose image is η . Let $U \subseteq X$ be a non-empty open subscheme. Let $x \in X \setminus U$ be a point. Let I_x be the inertia subgroup (Definition 2.0.3) of $\pi_1^{\text{ét}}(U, \bar{\eta})$ associated to x , well defined up to conjugation.

Let Λ be a finite ring and let $\mathcal{F}$ be a locally constant sheaf of finitely generated Λ -modules on $U_{\text{ét}}$. The inertia subgroup I_x acts continuously on the stalk $\mathcal{F}_{\bar{\eta}}$, which is a finitely generated Λ -module. For any field k , let $V = \mathcal{F}_{\bar{\eta}} \otimes_{\Lambda} k$ be the resulting finite-dimensional k -vector space with induced I_x -action.

The *drop of $\mathcal{F}$ at x* is the drop of the representation V under the action of I_x :

$$\text{drop}_x(\mathcal{F}) = \dim_k(V) - \dim_k(V^{I_x}).$$

When Λ is a field, this simplifies to $\dim_{\Lambda}(\mathcal{F}_{\bar{\eta}}) - \dim_{\Lambda}(\mathcal{F}_{\bar{\eta}}^{I_x})$.

For a lisse R -sheaf $\mathcal{F} = (\mathcal{F}_n, u_n)_{n \geq 1}$ on $U_{\text{ét}}$ where (R, λ) is a complete DVR with fraction field E , the *drop of $\mathcal{F}$ at x* is defined as the drop of the E -vector space $\mathcal{F}_{\bar{\eta}} \otimes_R E$ under the induced I_x -action.

The drop is independent of the choice of valuation centered at x because different choices yield conjugate inertia subgroups, and conjugation does not change the dimension of the invariant subspace. The drop vanishes if and only if $\mathcal{F}$ is unramified at x (Definition 2.0.4).